SWYNEX TECHNOLOGIES

EMPLOYEE ANALYTICS CASE STUDY

Final Data Analytics Project | Task 4

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Tools: Microsoft Excel • Power Query • SQL • Power BI

PROJECT REPORT

1. Executive Summary

This case study brings together the employee-data cleaning, exploratory data analysis (EDA), and Power BI dashboard work completed during the Swynex Technologies Data Analytics internship. The aim is to turn a raw employee dataset into a structured, readable view of workforce composition, salary patterns, work arrangements, and data-quality considerations.

The workflow used Microsoft Excel and Power Query for initial inspection and cleaning, SQL for structured analysis, and Power BI for visual reporting.

2. Problem Statement

The employee dataset contains workforce information such as department, region, employment status, salary, performance score, remote-work indicator, joining date, age, and contact details. Before this information can support business decisions, its quality must be checked and its key patterns summarized.

The project addresses the following questions:

- What is the overall distribution of employees by department, region, and employment status?
- How does average salary vary across departments?
- How are employees split between remote and non-remote work?
- What performance-score categories are represented in the dataset?
- Which data-quality issues should be considered before interpreting the results?

3. Dataset Overview

The dataset is an employee-level table with one record intended to represent one employee. The initial file contained 1,021 rows and 12 columns. After import into the SQL environment, the working table contained 1,020 records. This difference should be reconciled against the source file before the project is treated as a production report.

Field	Description	Analytical use
Employee_ID	Employee identifier	Record checks / distinct employee count
First_Name, Last_Name	Employee name fields	Basic completeness checks
Age	Employee age	Age analysis, subject to missing values
Department, Region	Organizational grouping	Headcount and salary comparisons
Status	Active / Inactive / Pending	Workforce status distribution
Salary	Employee salary	Average salary and department comparison
Email, Phone	Contact fields	Format and completeness checks
Performance_Score	Poor / Avg / Excellent	Performance category distribution
Remote_Work	True / False indicator	Remote-work split
Join_Date	Employee joining date	Tenure or joining-trend analysis if validated

4. Tools and Workflow

- **Microsoft Excel:** Initial dataset inspection, review of missing values, and supporting pivot tables/charts.
- **Power Query:** Data profiling, type correction, text/date cleanup, and preparation before loading.
- **SQL (PostgreSQL):** Record checks, grouped summaries, and exploratory analysis using SELECT, WHERE, GROUP BY, and aggregate functions.
- **Power BI:** Interactive dashboard to communicate workforce KPIs and comparisons visually.

5. Data Cleaning and Preparation

The following checks and transformations were carried out during preparation:

- Checked Employee_ID for duplicates; no duplicate IDs were found in the reviewed data.
- Reviewed missing values. Age had 209 missing rows; salary had 24 null/empty values in the later cleaning check.
- Standardized the Join_Date format where separators were inconsistent.
- Reviewed phone values and removed hyphens; phone was changed to text because it is an identifier, not a numeric measure.
- Flagged invalid phone values, including negative values and entries that did not meet the expected 10-digit format. The review noted 92 invalid or non-10-digit rows.
- Reviewed salary and age completeness before interpreting related metrics.
- Split combined Department-Region information where required so the fields could be analyzed separately.
- Loaded the prepared data into the analysis environment and checked the imported row count.

6. Exploratory Data Analysis

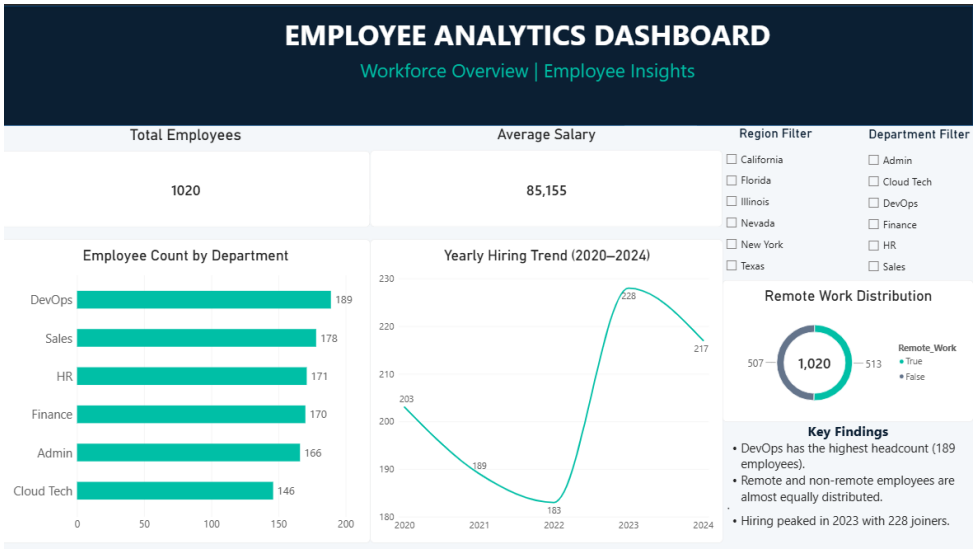
SQL was used to summarize employee records and explore distributions. Grouped queries helped compare departments and statuses, while aggregate functions were used to calculate counts and average salary.

Analysis question	Observed result / method	Interpretation
Employee count by status	Pending: 356; Active: 352; Inactive: 312	Pending is the largest of the three status groups in the imported table.
Remote-work distribution	True: 513; False: 507	The dataset is almost evenly split between remote and non-remote indicators.
Employees in Sales	178 employees	Sales is a sizeable department in the current dataset.
Average salary in Sales	Approximately 86,896	Confirm salary currency/units and null-handling before external interpretation.
Missing phone values	92 rows with Phone IS NULL	Contact completeness is a data-quality consideration.
Missing salary values	24 rows with Salary IS NULL	Salary comparisons should account for missing values.
Duplicate employee IDs	No duplicates found in the reviewed check	The identifier check did not flag repeated Employee_ID values.

7. Dashboard Design

The Power BI dashboard uses a dark navy and teal visual theme to create contrast and maintain a consistent visual hierarchy. KPI cards provide quick summaries, while charts support comparison across employee groups.

- KPI cards: use for headline metrics such as employee count and average salary, ensuring the aggregation is correct.
- Department comparison: use a bar or column chart to compare headcount or average salary by department.
- Employment status: use a bar or donut chart to show Active, Inactive, and Pending categories.
- Remote work: use a simple comparison visual for True vs. False.
- Performance score: show the distribution of Poor, Avg, and Excellent categories.
- Slicers: allow users to filter the report by department, region, or status where these fields are available.



8. Key Business Insights

- Employment status is distributed across all three categories, with Pending (356) slightly higher than Active (352) and Inactive (312). The business may want to clarify what 'Pending' means operationally.
- Remote-work indicators are nearly balanced (513 True and 507 False), suggesting both work arrangements are represented in similar proportions in this dataset.
- Sales has 178 employees in the imported table. Department-level comparisons can help stakeholders understand workforce allocation.
- The average Sales salary is approximately 86,896 in the dataset; the currency and salary period should be documented before making compensation decisions.
- Missing salary and phone values, missing ages, and invalid phone formats demonstrate why data-quality checks are necessary before drawing conclusions.

9. Conclusion

This project demonstrates an end-to-end introductory analytics workflow: inspecting and cleaning a dataset, using SQL to answer structured questions, and presenting findings through a Power BI dashboard. The analysis shows how clear metrics and visual summaries can make employee data easier to interpret, while the cleaning notes highlight the importance of validating source data and documenting limitations.

Example SQL Queries

1. Count employees by status

```
SELECT Status, COUNT(*) AS employee_count
FROM employees
GROUP BY Status
ORDER BY employee_count DESC;
```

2. Remote-work distribution

```
SELECT Remote_Work, COUNT(*) AS employee_count
FROM employees
GROUP BY Remote_Work;
```

3. Headcount by department

```
SELECT Department, COUNT(*) AS employee_count
FROM employees
GROUP BY Department
ORDER BY employee_count DESC;
```

4. Average salary by department

```
SELECT Department, AVG(Salary) AS average_salary
FROM employees
WHERE Salary IS NOT NULL
GROUP BY Department
ORDER BY average_salary DESC;
```

5. Find missing salary values

```
SELECT COUNT(*) AS missing_salary_rows
FROM employees
WHERE Salary IS NULL;
```

6. Check duplicate employee IDs

```
SELECT Employee_ID, COUNT(*) AS id_count
FROM employees
GROUP BY Employee_ID
HAVING COUNT(*) > 1;
```